

Final Report

Introduction

Housing prices are influenced by a combination of structural characteristics, property condition, amenities, and time-related housing factors rather than by any single variable alone. Because of this, we apply multiple linear regression, which is useful both for prediction and for understanding how different predictors are associated with sale price. The purpose of this project is to develop a multiple linear regression model that predicts residential sale prices using selected housing characteristics from the training dataset.

Multiple linear regression was chosen because it is transparent, interpretable, and appropriate for studying how several predictors jointly relate to a continuous response variable. Unlike more complex machine learning methods, regression allows direct interpretation of coefficients, assessment of statistical significance, and formal evaluation of model assumptions. This is especially important for this project because the goal is not only to obtain a competitive Kaggle score but also to justify the validity and interpretability of the final model.

The analysis proceeds in several stages. First, the data and selected predictors are described through summary statistics and graphical exploration. Second, an initial full model using the original variables is fitted and assessed through diagnostic plots. Third, transformations are considered to improve model assumptions. Finally, variable selection is used to obtain a more parsimonious model, and the resulting final model is interpreted in the context of housing prices.

Data Description

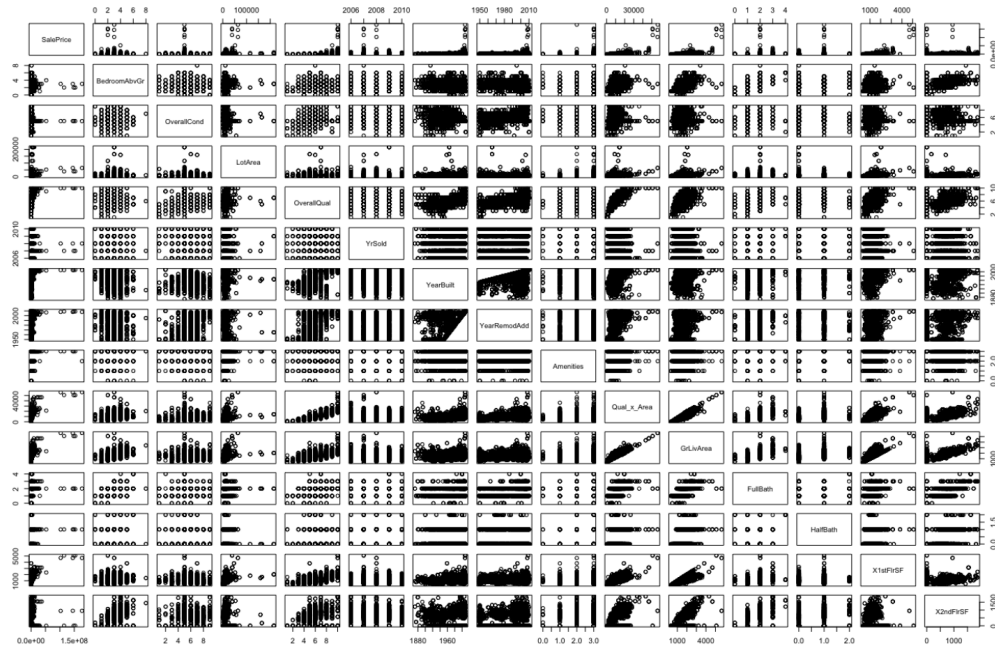
The training dataset contains 7,000 observations, and the test dataset contains 3,000 observations. Initially, there were 89 variables to work with, after doing some research, we chose a select few that we felt made the strongest impact on sales price to use for our model. The response variable is SalePrice, representing the sale price of a residential property. For our initial model, the predictors considered were BedroomAbvGr, OverallCond, LotArea, OverallQual, YrSold, YearBuilt, YearRemodAdd, Amenities, Qual_x_Area, GrLivArea, FullBath, HalfBath, X1stFlrSF, and X2ndFlrSF. The variable Qual_x_Area was constructed as the product of OverallQual and GrLivArea in order to capture the combined effect of home quality and living area. The variable amenities were created as the sum of four binary indicators representing the presence of a garage, central AC/heater, basement, and fireplace.

The predictors in the model include several different data types. Variables such as LotArea, GrLivArea, X1stFlrSF, and X2ndFlrSF are quantitative size-related predictors. Variables such as Amenities, BedroomAbvGr, FullBath, and HalfBath are count-based predictors. OverallCond and OverallQual act as ordinal rating variables, while YrSold, YearBuilt, and YearRemodAdd are temporal predictors. The engineered predictor Amenities was included to summarize housing features and possible interaction effects.

To explore the relationships among variables, a scatterplot matrix of the initial predictors was created. The plot showed that several size-related predictors were positively associated with one another. In particular, Qual_x_Area displayed a strong relationship with both OverallQual and GrLivArea, which is expected because it is defined from those variables. This suggested possible multicollinearity in the full model.

Several variables such as BedroomAbvGr, FullBath, HalfBath, Amenities, and YrSold appeared discrete, producing vertical banding in the scatter plots rather than continuous point clouds. Most importantly, the relationships between SalePrice and the predictors did not appear strongly linear on the original scale, and the response variable appeared right-skewed. These observations suggested that transformation might improve the regression fit.

Matrix Scatter Plot of Original Data



Summary of the full model

```
lm(formula = cleanhouse)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-25509265 -764353  64694 1015698 127044368
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.312e+07  6.230e+07  -0.853  0.393929
BedroomAbvGr -2.336e+05  6.404e+04  -3.648  0.000266 ***
OverallCond   9.672e+04  4.473e+04   2.162  0.030629 *
LotArea      1.738e+01  4.399e+00   3.950  7.88e-05 ***
OverallQual  -3.051e+06  8.343e+04  -36.565 < 2e-16 ***
YrSold       2.802e+04  3.099e+04   0.904  0.365885
YearBuilt    1.821e+03  2.418e+03   0.753  0.451307
YearRemodAdd 3.732e+03  2.890e+03   1.291  0.196590
Amenities    1.426e+05  7.241e+04   1.969  0.048973 *
Qual_x_Area  2.053e+03  4.737e+01  43.346 < 2e-16 ***
GrLivArea   -1.044e+04  7.765e+02  -13.445 < 2e-16 ***
FullBath    -4.975e+05  1.194e+05  -4.167  3.12e-05 ***
HalfBath    1.914e+05  1.192e+05   1.605  0.108460
X1stFlrSF   2.098e+03  7.503e+02   2.796  0.005191 **
X2ndFlrSF  -1.447e+03  7.533e+02  -1.921  0.054716 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 3413000 on 6985 degrees of freedom
Multiple R-squared:  0.432,    Adjusted R-squared:  0.4309
F-statistic: 379.5 on 14 and 6985 DF, p-value: < 2.2e-16
```

Analysis of Variance Table

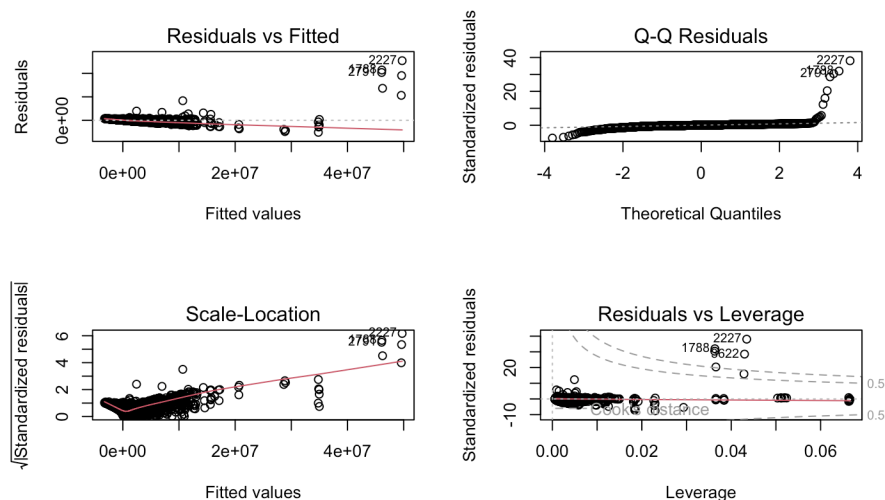
```
Response: SalePrice
            Df    Sum Sq   Mean Sq    F value    Pr(>F)
BedroomAbvGr 1 1.1534e+13 1.1534e+13    0.9902  0.319727
OverallCond  1 7.2932e+14 7.2932e+14   62.6152 2.902e-15 ***
LotArea      1 5.8608e+15 5.8608e+15  503.1784 < 2.2e-16 ***
OverallQual  1 9.7493e+15 9.7493e+15  837.0219 < 2.2e-16 ***
YrSold       1 4.1260e+11 4.1260e+11   0.0354  0.850716
YearBuilt    1 9.9017e+13 9.9017e+13   8.5011  0.003561 **
YearRemodAdd 1 4.5853e+11 4.5853e+11   0.0394  0.842730
Amenities    1 6.8990e+11 6.8990e+11   0.0592  0.807723
Qual_x_Area  1 2.6510e+16 2.6510e+16 2275.9784 < 2.2e-16 ***
GrLivArea    1 1.1391e+16 1.1391e+16  977.9607 < 2.2e-16 ***
FullBath     1 3.4729e+14 3.4729e+14  29.8163 4.913e-08 ***
HalfBath     1 1.8725e+15 1.8725e+15 160.7653 < 2.2e-16 ***
X1stFlrSF    1 5.2681e+15 5.2681e+15  452.2888 < 2.2e-16 ***
X2ndFlrSF    1 4.3003e+13 4.3003e+13   3.6920  0.054716 .
Residuals   6985 8.1359e+16 1.1648e+13
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$R^2 : 0.432$; Overall P-Value of Anova:< 2.2e-16

Results & Interpretation

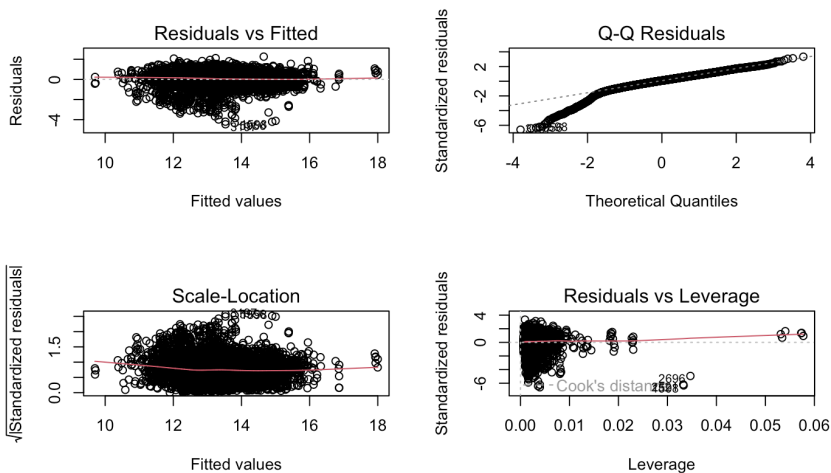
In order to analyze the relationship between the Sale Prices (Y) and the Housing Characteristics/Predictors (X), we made several regression models that we analyzed and compared at the end. Before we built our first finalized model, we briefly looked through the variables, picking the ones that were most interesting to us, and did preliminary research of the housing price predictors from the general housing market using the internet to choose a smaller set of meaningful variables. The models that we discuss below are the main steps to our final model, although tested alternative transformations and explored more model variations during the process. The main purpose of doing this was to identify a specific model that can explain the variation of housing prices, which also fulfills the assumptions of linear regression.

The first model that we tested is the full model, named 'm1 full'. This initial model used the default response variable SalePrice, with predictors: BedroomAbvGr, OverallCond, LotArea, OverallQual, YrSold, YearBuilt, YearRemodAdd, Amenities, Qual_x_Area, GrLivArea, FullBath, HalfBath, X1stFlrSF, and X2ndFlrSF. This model resulted in an adjusted R^2 of 0.431, which means that 43.1% of the variability can be explained by the predictors (x variables) in the model. From the ANOVA test, the p-value is less than 0.05 ($p < 2.2e-16$), indicating that the regression is highly significant, meaning that at least one of the predictors is correlated with the SalesPrice (Y). However, when we look at the diagnostic plot of this model, it showed several violations of the linear assumptions. For the Residuals vs. Fitted plot it showed that the residuals follow and spread out more as the fitted values get higher, which suggests heteroscedasticity. From the normal Q-Q plot, it also showed that the residuals are not normally distributed, shown by the upper tail. From the Scale-Location plot, it revealed non-constant variance. From the Residuals vs. Leverage plot, we spotted some influential points, but most of the observations are still inside the Cook's Distance range. Overall, the original model was statistically significant, but it did not satisfy all the regression assumptions to be valid.



To improve the model, we applied the Box-Cox method to apply power transformation to some of the selected variables as well as our response variable simultaneously. From the transformation results, it suggests the log transformation for the variables of SalePrice, LotArea, and also X1stFlrSF, while keeping the remaining variables without transformation. After the transformation was applied, the overall model

had a significant improvement. The adjusted R^2 squared value of this new model became approximately 0.641, meaning that 64% of the variation in housing prices is explained by the x variables/predictors. Next, by seeing the diagnostic plots, we can also see some improvements. The residuals become scattered around 0, it also had a more consistent spread across fitted values. While the Q-Q plot showed that the residuals were closer to normal distribution. This means that linearity, constant variance, and normality has been improved. From these improvements, we can tell that the transformation is now a valid transformation.



Although we saw improvements to the linear assumptions already through the transformation before, we continue to test the multicollinearity using the Variance Inflation Factor (VIF). Based on the test, some predictors showed VIF values greater than 5, meaning that multicollinearity exists among some variables. That is why we need to continue our work on the Variable Selection Model, and we ended up choosing the forward stepwise regression based on the Akaike Information Criterion (AIC) comparison. In addition to forward selection, we tested our model with other methods such as backward selection and all-subsets selection by comparing their AIC and AICc criteria. From this approach, the results were the exact same, so for simplicity, we present the forward stepwise model that we called m3forward as our final model representation. This method began with an intercept-only model, while we added the predictors sequentially that most improved the AIC value. For the resulting model, m3forward, includes predictors such as OverallQual, log(X1stFlrSF), log(LotArea), Amenities, BedroomAbvGr, YearBuilt, FullBath, Qual_x_Area, X2ndFlrSF, YearRemodAdd, OverallCond, GrLivArea, and HalfBath. Through this forward selection method, it balances the predictive performance and interpretability by removing unnecessary predictors and keeping the ones that give meaningful information to the model.

vif(m2)						

BedroomAbvGr	OverallCond	log(LotArea)	OverallQual	YrSold	YearBuilt	YearRemodAdd
1.772139	1.447062	1.489533	9.597595	1.006652	3.222722	2.206969
Amenities	Qual_x_Area	GrLivArea	FullBath	HalfBath	log(X1stFlrSF)	X2ndFlrSF
1.013775	38.770193	56.685941	2.604007	2.168186	15.665530	18.332989

```
vif(m3forward)
[[1]]
```

OverallQual	log(X1stFlrSF)	log(LotArea)	Amenities	BedroomAbvGr	YearBuilt	FullBath
9.595780	15.662633	1.489281	1.013547	1.770834	3.222354	2.603972
Qual_x_Area	X2ndFlrSF	YearRemodAdd	OverallCond	GrLivArea	HalfBath	
38.722955	18.325325	2.203915	1.446727	56.624922	2.168185	

The images above present the variable inflation (VIF) tables for both the m2 and m3forward models to illustrate the impact of our variable selection process and show the difference between the two models. As we observe the VIF table, we can see that the new model, m3forward, shows a slight improvement in VIF. For instance, the Qual_x_Area predictor had decreased by 0.047238 (38.770193 to 38.722955) from the previous model, m2. Although the VIF has only slightly decreased for the variables, it is important to note that some predictors have VIFs that remain greater than 5, confirming that the multicollinearity is still persisting. We expect multicollinearity between the Quality and FlrSF predictors, but removing the YrSold predictor still resulted in a slight improvement in comparison to the m2 model.

While we later tested models with other variables and other transformed models, among all of the candidate models that we constructed and tested, the best predictive model is the log-transformed regression model (m2) with forward stepwise variable selection. We found this model the best because the transformation has proven to improve the regression assumption, specifically for the normality and constant variance. The second reason is because the model explains a bigger proportion of variability in housing prices, and the last important reason is that the stepwise regression creates a simpler model while keeping the variables that have the most impact. Overall, the key predictors for this final model are housing quality, living area, lot size, construction year, remodeling year, and availability. amenities. The final model shows these relationships and provides a reliable way for predicting housing prices.

Discussion

This project showed that multiple linear regression can be used to meaningfully predict housing sale prices when the predictors are carefully selected and model assumptions are addressed. The initial full model on the original price scale was statistically significant, but the diagnostic plots showed that the assumptions of constant variance and normality were not satisfied. After examining the predictor relationships and applying Box-Cox-guided transformations, a log-transformed regression model provided a much better fit and more reasonable diagnostics.

The final model makes sense in a real-world housing context. Variables measuring house quality, living area, lot size, build year, remodeling recency, and amenities are all logically connected to market value. In practice, buyers are willing to pay more for larger, newer, higher-quality homes with desirable features. Therefore, the positive effects observed for several of these predictors align well with general housing market expectations. At the same time, some counterintuitive coefficients, such as the negative coefficient on OverallCond, likely reflect multicollinearity and the fact that partial regression coefficients measure the effect of a variable after controlling for all other predictors.

Several limitations should be acknowledged. First, the model considered only a subset of the 88 available predictors, so additional useful information may have been omitted. Second, some predictors were engineered in relatively simple ways, such as the Amenities count, which may not capture the full complexity of housing features. Third, while the transformed model improved diagnostics, some influential observations and multicollinearity may still remain. Finally, stepwise selection based on AIC can be useful for parsimony, but it is not guaranteed to find the single best model in every setting.

Future improvement could include more careful handling of categorical predictors such as neighborhood or housing style, consideration of additional engineered variables such as house age or total living area, and comparison of several candidate regression models using adjusted R squared, AIC, BIC, and out-of-sample prediction results together. Within the constraints of this project, however, the final transformed multiple linear regression model provided a reasonable balance of interpretability, validity, and predictive performance.

Challenges Encountered

One challenge in this project was deciding which predictors to include from the large pool of available variables. Because the dataset contained 88 potential predictors, it was necessary to balance completeness with simplicity and statistical validity. Another challenge was handling variables of different types, especially the presence of discrete, ordinal, and engineered predictors in the same model. In addition, the original full model clearly violated important regression assumptions, which required transformation of the response and selected predictors. Identifying meaningful transformations and then interpreting the resulting coefficients on the log scale required additional care. These challenges made the project more involved, but they also demonstrated the importance of statistical reasoning in building a defensible regression model.